

CHARLOTTE-MONROE, NC, USA

WMO: 723194

Lat: 35.017N

Lon: 80.621W

Elev: 679

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53872

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.8	24.9	1.8	6.2	27.9	6.8	8.0	31.6	22.2	49.8	19.7	49.3	4.8	30	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	94.8	74.2	92.4	74.0	90.3	73.5	77.4	88.5	76.4	87.0	75.5	85.4	7.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	132.1	81.1	73.3	127.1	80.0	72.6	123.9	79.4	41.1	88.3	40.2	87.1	39.3	85.7	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.5	14.2	DB	14.3	98.1	4.0	3.0	11.5	100.3	9.1	102.1	6.9	103.8	4.0	106.0	
			WB	12.1	79.1	3.6	1.6	9.6	80.3	7.5	81.2	5.5	82.1	2.9	83.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	42.4	45.6	52.9	61.7	69.3	77.0	79.7	78.6	73.1	62.2	51.8	45.9
	DBStd	15.16	10.32	9.17	9.28	7.70	6.43	4.45	3.55	4.12	5.50	7.85	8.34	8.99
	HDD50	826	282	180	80	8	0	0	0	0	0	7	76	193
	HDD65	2973	701	545	388	151	37	1	0	0	9	146	400	595
	CDD50	5120	47	57	168	358	600	809	922	886	693	384	131	65
	CDD65	1791	1	2	12	51	172	359	457	421	251	58	5	2
	CDH74	15692	3	10	99	484	1390	3273	4343	3749	1881	428	28	4
	CDH80	6406	0	0	9	96	432	1413	2025	1652	685	93	1	0
Wind	WSAvg	5.8	6.5	6.9	7.0	6.9	5.8	5.2	5.0	4.4	5.3	5.5	5.7	5.9
Precipitation	PrecAvg	44.0	3.5	3.0	4.0	3.5	3.2	4.2	4.6	4.1	3.9	3.4	3.1	3.6
	PrecMax	59.7	6.8	5.7	8.3	8.7	9.1	7.2	9.3	9.5	9.3	14.4	9.3	8.7
	PrecMin	30.6	1.0	1.1	1.3	0.9	0.1	1.2	1.9	0.6	0.3	0.0	0.3	1.5
	PrecStd	6.7	1.5	1.1	1.8	1.9	1.9	1.9	1.7	2.0	1.9	2.8	2.1	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	75.6	81.2	86.9	91.1	96.7	97.5	98.6	93.1	87.2	77.4	73.2
		MCWB	61.4	63.0	62.6	66.5	71.6	73.8	75.5	73.7	71.7	71.6	62.9	63.4
	2%	DB	66.8	69.8	76.8	82.6	87.8	93.1	95.0	93.6	90.2	82.1	73.2	68.6
		MCWB	58.9	58.6	61.0	64.7	70.0	73.8	75.3	73.9	71.4	68.1	61.3	62.3
	5%	DB	63.2	65.6	72.7	79.8	84.8	90.5	92.4	91.1	87.4	79.0	70.0	64.9
		MCWB	56.5	56.8	59.0	63.3	69.2	73.1	74.8	74.3	71.3	66.2	61.1	59.9
	10%	DB	58.9	61.4	68.5	76.2	81.8	88.0	90.1	88.5	84.0	75.1	66.2	61.4
		MCWB	52.3	53.0	56.4	61.9	68.2	72.3	74.4	73.8	70.5	65.4	58.5	56.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	65.4	65.4	70.9	74.8	77.1	78.8	78.7	76.3	75.2	68.8	67.3
		MCDB	69.2	71.1	75.3	81.3	84.2	89.4	91.0	88.3	85.4	80.7	71.8	69.8
	2%	WB	61.3	62.2	63.5	68.4	72.7	75.8	77.6	77.3	74.9	72.4	65.6	63.9
		MCDB	65.9	67.4	71.8	77.1	82.2	87.9	89.6	87.2	83.6	77.9	69.5	67.2
	5%	WB	58.1	58.6	61.7	66.3	71.3	74.9	76.7	76.3	73.6	69.8	63.4	60.9
		MCDB	62.4	64.3	70.0	74.7	80.7	86.4	88.0	85.7	81.9	75.2	67.4	64.3
	10%	WB	53.6	54.1	59.4	64.3	69.9	73.8	75.7	75.2	72.4	67.2	60.2	57.2
		MCDB	57.7	58.8	65.7	72.5	78.4	84.5	86.2	84.2	79.9	73.6	65.4	60.7
Mean Daily Temperature Range	5% DB	MDBR	18.8	19.4	20.9	21.9	20.4	19.7	19.5	18.5	18.6	20.5	21.3	18.5
		MCDBR	21.8	23.2	24.5	25.2	23.5	22.9	22.8	22.1	22.9	23.4	23.7	20.3
		MCWBR	15.1	14.5	12.1	11.2	9.3	7.6	6.9	6.7	7.8	10.7	14.0	14.2
	5% WB	MCDBR	18.7	20.6	20.9	20.8	19.5	20.2	19.9	18.3	18.1	17.0	17.8	17.5
	MCWBR	15.1	15.5	12.2	11.0	8.5	7.6	7.1	6.8	7.2	9.5	13.0	14.4	
Clear-Sky Solar Irradiance	taub	0.312	0.322	0.358	0.403	0.460	0.486	0.539	0.515	0.431	0.381	0.330	0.310	
	taud	2.504	2.473	2.402	2.264	2.149	2.115	1.991	2.045	2.267	2.384	2.487	2.544	
	Ebn at Noon	284	293	290	280	264	256	242	245	262	268	274	278	
	Edh at Noon	26	30	35	43	49	50	56	53	40	33	27	24	
All-Sky Solar Radiation	RadAvg	844	1049	1391	1741	1927	2043	1916	1747	1486	1223	930	729	
	RadStd	49	121	101	134	157	145	122	82	129	152	75	66	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+360	+261
(47)	Regional (71 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+297	+207

Nomenclature: See separate page